

Exploring ~~LLMs for EEG Classification through~~ ~~Active Prompting Decoding~~ with ~~LLMs: A~~ Purity-Guided ~~Unique-Round Example Selection~~ (~~PURE~~)~~Active Prompting Framework~~

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Abstract—Large language models (LLMs) have demonstrated impressive in-context learning (ICL) capabilities, particularly in few-shot prompting scenarios. However, most studies have ~~focused-concentrated~~ on natural language processing (NLP) and computer vision (CV), with ~~limited-exploration-of-tasks based-on-minimal exploration into domains involving~~ complex physiologic signals such as electroencephalogram (EEG) ~~signal classification decoding~~. To bridge this gap, we propose a purity-guided active prompting ~~approach~~(PGAP) framework, which integrates domain-specific EEG feature extraction with an active learning-based demonstration selection strategy. ~~PURE-PGAP~~ identifies a compact set of prototypical examples that effectively activate the few-shot capabilities of LLMs without requiring gradient-based parameter updates, and is compatible with both open-source and closed-source models. Unlike the conventional framework that involves random ~~selection-of-demonstration examples demonstration selection~~ on a per-instance basis for each test sample, ~~PURE actively identifies a small set PGAP performs a one-time identification~~ of highly representative EEG samples ~~just once~~ from the training ~~dataset~~, significantly reducing computational overhead. Extensive experiments on seven EEG datasets and three paradigms demonstrate that ~~PURE consistently outperforms random selection baseline PGAP consistently outperforms five existing sample selection baselines~~, achieving superior accuracy and robustness. Furthermore, performance improves with more powerful LLMs (e.g., DeepSeek-V3), and in some cases, even surpasses state-of-the-art supervised models.

Index Terms—Brain-computer interface, LLMs, active learning, prompt engineering, in-context learning

I. INTRODUCTION

Transformer-based large language models (LLMs) [1] have achieved remarkable progress across ~~various a wide range of~~ natural language processing (NLP) tasks. As model ~~size scale~~ increases, LLMs exhibit strong in-context learning (ICL) capabilities, enabling them to perform downstream tasks by conditioning on a few demonstration examples during inference, without any ~~gradient-based~~ parameter updates [2].

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Brain-computer interfaces (BCIs) [3], [4] enable direct communication between the human brain and external devices, allowing users to interact with environment through brain signals that reflect cognitive states or intentions [5]. Among BCI modalities, electroencephalography (EEG) is the most widely adopted due to its low cost, non-invasiveness, and ease of acquisition.

~~While-Despite~~ ICL has been extensively studied in ~~natural language processing (NLP) NLP~~ and computer vision (CV), its application in BCI tasks, especially ~~those~~-based on EEG signals, remains underexplored. To bridge this gap, it is essential to ~~develop effective strategies for embedding complex numerical data-exploit strategies that effectively embed high-dimensional physiological signals~~ into textual prompts ~~that can be processed by LLMs suitable for LLM processing~~.

Prior efforts have investigated such integration in domains such as tabular data [6], [7] and medical records [8], where numerical values often possess clear semantic meaning and can be easily verbalized. ~~In-contrast~~However, EEG signals lack ~~inherent semantic interpretability. For example such intrinsic semantic context. For instance,~~ the value “16” in a clinical record may represent a patient’s age, while in EEG data it may simply denote a ~~signal-voltage~~ amplitude at a specific time point, devoid of symbolic meaning.

~~Moreover, EEG signals are high-dimensional and temporally dense. Inputting The high temporal density and dimensionality of EEG signals further complicate the usage of LLMs. Directly feeding such dense and unstructured data into an LLM results in may lead to excessive token consumption and limits, thereby limiting the model’s capacity to extract meaningful patterns. Additionally Moreover,~~ EEG signals are ~~inherently~~ non-stationary, exhibit substantial ~~inter-subject-cross-subject~~ variability, and suffer from low signal-to-noise ratios, posing ~~significant~~ challenges for direct LLM-based analysis. ~~An overview of LLM applications across various domains, with a particular emphasis on the challenges of EEG decoding, is shown in Fig. ??.~~

To address these challenges, we propose a simple yet effective ~~purity-based active prompting (PGAP)~~ framework that leverages ~~domain-domain-specific~~ knowledge to extract compact and discriminative EEG ~~features-representations~~ suitable for LLM processing. ~~We also Specifically, we~~ introduce a purity-based example selection strategy ~~, which that~~ identifies

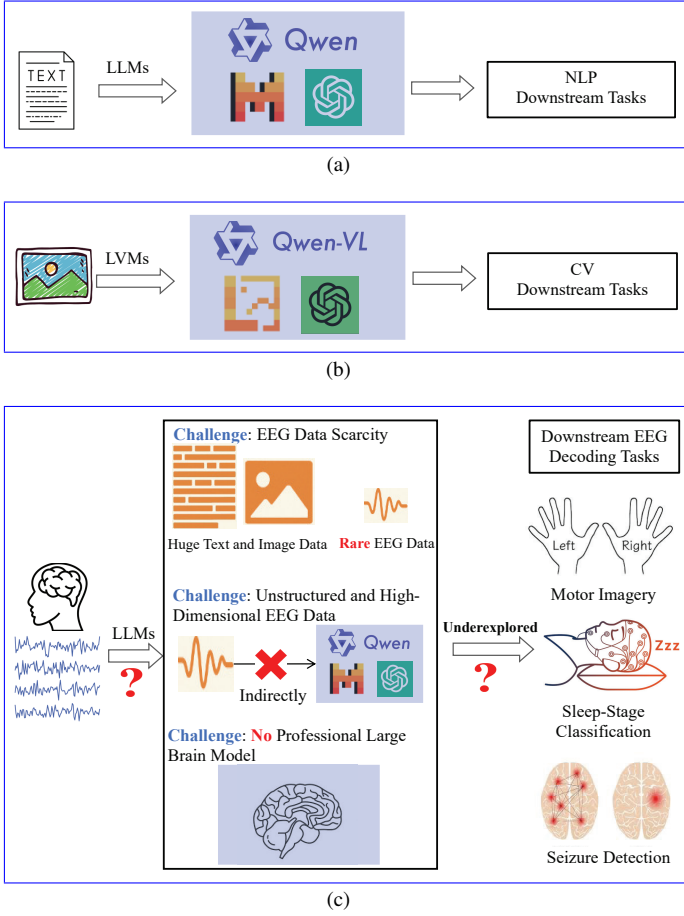


Fig. 1. LLM applications across various domains. (a) NLP; (b) CV; and (c) EEG decoding.

highly consistent intra-class samples based on the k -nearest neighbor purity criterion. Experimental results on seven EEG datasets across three paradigms demonstrate that PURE-PGAP consistently outperforms existing sample selection algorithms.

In summary, our contributions in this paper are main contributions are as follows:

- 1) To the best of our knowledge, this work is the first to directly explore the feasibility of using LLMs for classifying EEG signals via attempt to harness LLMs for EEG signal classification through in-context few-shot prompting, thereby extending the application applicability of LLMs to complex, low-semantic numerical domains beyond natural language physiological domains beyond traditional text and vision tasks.
- 2) We incorporate domain-specific knowledge to extract features from EEG signals. This not only reduces the number of input tokens substantially but also improves classification performance, making neuroscience-informed feature extraction to reduce the input dimensionality and token count, enhancing both computational efficiency and downstream performance within the LLM-based prompting pipeline more effective and computationally feasible.
- 3) We introduce a one-time propose a one-shot ex-

ample selection strategy that selects—identifies a fixed set of representative EEG samples from the training set and reuses them across all test instances, substantially—significantly reducing inference cost and latency compared to conventional per-instance selection approaches commonly used in NLP prompting approaches.

- 4) Experiments on seven EEG datasets and three paradigms demonstrate that PURE-PGAP consistently outperforms existing sample selection approaches in both accuracy and robustness.

The remainder of this paper is organized as follows: Section II introduces related work. Section III proposes PURE details the proposed PGAP. Section ?? introduces datasets. Section IV presents the experimental results. Finally, Section V draws conclusions.

II. RELATED WORKS

This section introduces related works on EEG-based classification of EEG decoding, active learning, and demonstration selection for ICL.

A. EEG-based Classification EEG Decoding with Machine Learning

EEG-based classification EEG decoding encompasses a wide range of tasks, including motor imagery (MI) [9], [10], sleep stage classification [11], and epilepsy detection [12] [13]. Traditional approaches typically rely on feature extraction methods approaches such as common spatial pattern (CSP) [14], frequency-domain features (e.g., power spectral density), time-domain features (e.g., amplitude statistics, zero-crossing rate), and nonlinear entropy features (e.g., entropy, complexity measures) [15], followed by classifiers such as logistic regression (LR) [16] or support vector machines (SVM) [17], etc.

To enhance EEG feature extraction, Ang *et al.* [18] proposed the filter bank CSP (FBCSP) method approach, which applies CSP to multiple frequency bands and automatically selects the most discriminative features for classification. Lotte *et al.* [19] introduced regularized CSP to improve the robustness of feature extraction.

Recent advances in deep learning have enabled end-to-end models that jointly perform feature extraction and classification, achieving strong performance in EEG decoding tasks [20], [21]. In particular, convolutional neural networks (CNNs) have demonstrated superior performance in MI classification. Schirrneister *et al.* [22] proposed ShallowCNN, which uses temporal and spatial convolutional filters to extract relevant features discriminative representations for MI decoding. Lawhern *et al.* [23] introduced EEGNet, a compact model employing depthwise and separable convolutions to significantly reduce the number of parameters while achieving strong performance across various BCI tasks, including MI classification. Song *et al.* [24] proposed EEGConformer that sequentially integrates ShallowCNN with a Transformer block, facilitating high-level feature extraction. Wang *et al.* [25] proposed MVCNet, a dual-branch network that parallelly

integrates CNN and Transformer blocks to capture both local spatial-temporal features and global temporal dependencies, thereby enabling more informative feature extraction for MI decoding. Jiang *et al.* [26] proposed CSP-Nets, a hybrid model that integrates knowledge-driven CSP filters with data-driven CNNs to ~~enhance~~ improve MI classification performance.

More recently, Jiang *et al.* [27] proposed a 0.369B-parameter large-scale model trained on a diverse EEG corpus spanning multiple paradigms and tasks. However, due to the limited size of publicly available EEG datasets, model scalability remains constrained, and its performance and usability in downstream applications still fall short compared to LLMs. Additionally, due to the modality gap, no existing study has directly applied LLMs to EEG classification tasks via prompt-based ~~in-context learning~~ ICL.

B. Active Learning

Traditional ~~Active Learning~~ active learning (AL) algorithms primarily emphasized uncertainty-based sampling ~~methods~~ approaches, which prioritize samples near the decision boundary or those associated with high classification uncertainty [28]. Lewis *et al.* [29] demonstrated that uncertainty sampling significantly reduces annotation costs in text classification tasks, outperforming both random and relevance sampling ~~methods~~ approaches, particularly when training data is limited.

Beyond uncertainty, representativeness has emerged as a complementary criterion. Strategies such as clustering-based sampling [30] and density-weighted uncertainty sampling [31] seek to select samples that are both uncertain and representative, avoiding outliers.

Multi-criteria approaches further enhance AL performance. Shen *et al.* [32] applied informativeness, representativeness, and diversity jointly to named entity recognition. Huang *et al.* [33] introduced QUIRE, which selects samples by minimizing maximum model change (MMC), effectively balancing informativeness and representativeness. Cai *et al.* [34] proposed a MMC strategy tailored to SVM, although it is sensitive to initial parameters. He *et al.* [35] formulated a batch-mode AL ~~method~~ approach combining uncertainty, representativeness, and diversity. Their ~~method~~ approach quantifies informativeness as the product of uncertainty and representativeness, clusters samples via kernel k-means, and selects cluster centers for labeling.

In regression tasks, Burbidge *et al.* [36] applied a query-by-committee (QBC) strategy based on model disagreement, though inaccurate model assumptions may degrade performance. Wu *et al.* [37] proposed RD, a sequential pool-based approach that jointly considers representativeness and diversity. RD can be combined with existing regression-based AL ~~methods~~ approaches such as QBC, MCM, and Greedy ~~Sampling~~ sampling (GS) to further enhance performance. Wu *et al.* [38] further demonstrated that GS improves regression performance by maximizing sample diversity.

Despite its success in conventional machine learning, AL has been rarely applied to prompt selection in LLMs. Whether the selection criteria developed for conventional models re-

main effective in few-shot prompting scenarios with LLMs remains an open question.

C. Demonstration Selection for ~~In-Context Learning~~ ICL

~~With the scaling of model and data sizes, LLMs have demonstrated the ability to perform ICL,~~ a paradigm wherein models learn tasks from a few example demonstrations without parameter updates [39], [40].

With the scaling of model and data sizes, LLMs have demonstrated the ability to perform ICL. Most existing approaches to demonstration selection adopt a per-instance strategy, where demonstrations are selected independently for each test instance. This leads to two key selection objectives: similarity and diversity [41].

Liu *et al.* [42] proposed selecting demonstrations based on similarity by retrieving the k nearest neighbors of the test instance using sentence encoders in the embedding space. However, this approach does not consider diversity, which may result in homogeneous and redundant examples. Moreover, in classification tasks, k NN-based selection often biases the model toward predicting the majority label among the selected examples, effectively reducing the LLM to a k NN classifier and diminishing its unique generalization capabilities.

To address the issue of homogeneity, Li *et al.* [43] introduced clustering-based retrieval, where the sample pool is first clustered, and the nearest instance to the query is selected from each cluster. While this approach increases diversity, it may include examples that lack sufficient similarity to the query.

To better balance similarity and diversity, iterative retrieval ~~methods~~ approaches have been proposed. These ~~methods~~ approaches begin by selecting the example most similar to the query and then sequentially add examples by considering both their similarity to the query and their diversity relative to previously selected samples. Scarlatos *et al.* [44] trained an ~~LSTM-based long short-term memory (LSTM) based~~ retriever within a reinforcement learning framework. During inference, the retriever selects an initial demonstration by processing the query, then updates the query representation using the LSTM's hidden state to incorporate prior demonstrations. This iterative process continues until k demonstrations are selected [41].

In CV, similar principles have been applied by scoring candidates using similarity metrics (e.g., cosine distance) to quantify relevance for prompt inclusion [45].

In contrast to per-instance approaches, Hong *et al.* [46] proposed a two-step ~~method~~ approach: first, a diverse and representative subset is selected for annotation; then, task-specific demonstrations are retrieved from this labeled pool at inference.

Overall, existing work on ICL demonstration selection has primarily focused on NLP and CV tasks, with strategies largely centered on balancing similarity and diversity among examples. Designing principled selection criteria for complex, non-semantic inputs, like EEG, presents a significant challenge.

III. METHOD

This section introduces ~~our proposed method, PURE~~ the proposed PGAP. Specifically, we start with a large pool of

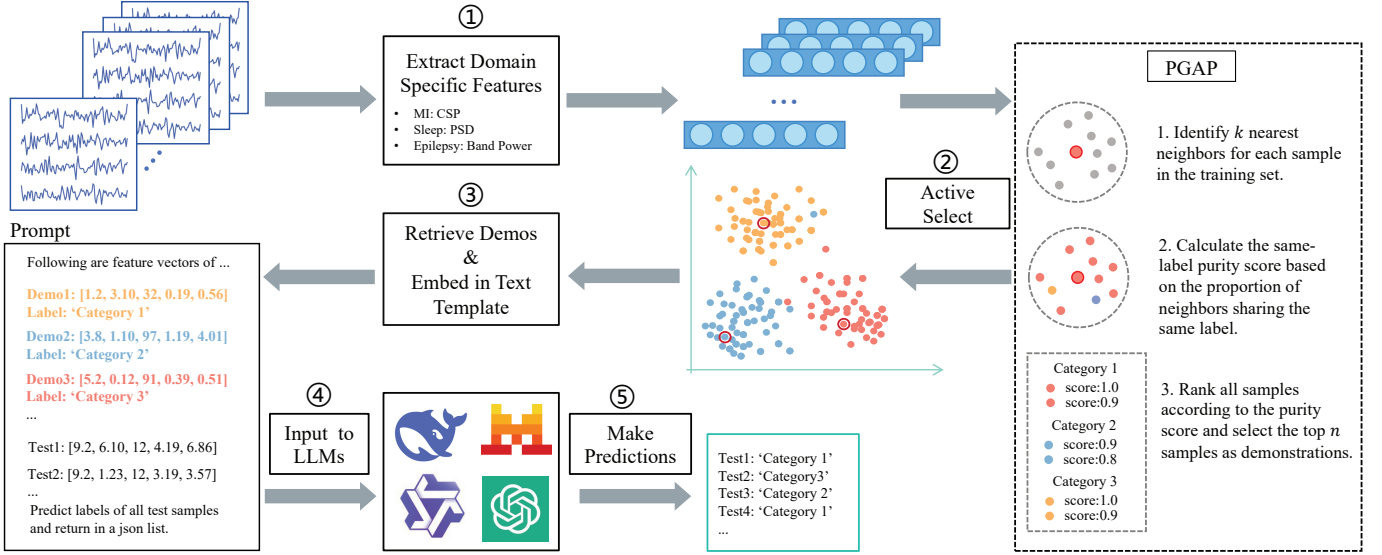


Fig. 2. Flowchart of EEG-based classification using LLM few-shot prompting.

labeled training samples and select a compact subset to serve as in-context examples within the prompt, aiming to fully leverage the few-shot capabilities of the LLM and maximize its performance on the test set. The overall procedure is illustrated in Figure 2.

A. Domain Knowledge-Driven Specific Feature Extraction

Due to the inherent complexity of raw EEG signals, it is necessary-essential to extract compact and discriminative handcrafted features after-following standard preprocessing. This step is crucial not only for enhancing task-relevant signal characteristics but also for reducing the input dimensionality to meet the token-length constraints of LLMs. The resulting embeddings serve-are subsequently utilized as demonstrations within the in-context prompt.

Domain-specific knowledge needs to be leveraged to design feature extraction pipelines tailored to different EEG paradigms. For MI-classification, which MI paradigm primarily elicits event-related desynchronization (ERD)-and synchronization (ERS)-and synchronization patterns in the μ (8-13 Hz) and β (13-30 Hz) frequency bands over the sensorimotor cortex, which-particularly around electrodes C3 and C4, we-We adopt a feature extraction pipeline inspired by the FBCSP framework [18] for MI EEG decoding. Specifically, signals are bandpass-filtered in the 8-24 Hz range, and divided into four 4-Hz sub-bands, and two-. Two CSP features are extracted from each band, resulting-in-yielding an 8-dimensional representation-feature vector per trial.

For sleep stage classification, EEG features are extracted from time, frequency, and time-frequency domains. In the time domain, standard statistical features such as amplitude measures and zero-crossing rates are employed. In the frequency domain, power spectral density and band power features are commonly used. Time-frequency domain approaches include features are obtained using approaches such as wavelet transform, short-time Fourier transform, empirical mode decomposition, and Wigner-Ville distribution. Since-Given that

our primary goal is to evaluate the impact of demonstration selection strategies (random vs. active), we focus on delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), spindle (12-14 Hz) and beta (14-30 Hz) bands, which have been shown to correspond to sleep stages W, N1-N3, and REM in prior EEG-literature, and the literature. We adopt a simple yet effective feature representation using-based on multi-band power spectral density features.

For epileptic seizure detection, previous studies have explored various domain features [47] [13], [47], highlighting the effectiveness of amplitude-based representations such as variance, energy, and entropy. Motivated by these insights, we adopt a practical and compact feature representation by computing band-specific signal power directly from raw EEG signals. Specifically, we apply bandpass filters to decompose the signals into canonical frequency bands— δ (0.5-4Hz), θ (4-8Hz), α (8-13Hz), β (13-30Hz), and γ (30-45Hz)—we then-extract-. Then, the average power within-is extracted from each band. This approach effectively captures band-level spectral energy pattern, which-is-patterns, which are highly indicative of seizure-related dynamics, while maintaining low computational complexity.

B. Purity-Based—Unique-Round—Example—Selection (PURE)PGAP

After obtaining compact feature vectors, we then-proceed to select a small subset of examples that can-most effectively activate the few-shot learning capabilities of the LLM. However, as mentioned-abovepreviously mentioned, it remains unclear whether principles from conventional AL are directly applicable to EEG-based classification-decoding with LLMs. To investigate-address this, we conducted a series of experiments to systematically evaluate the effectiveness of different example selection strategies.

Surprisingly, and contrary-to-conventional-assumptions, we-found-Contrary to prevailing assumptions, our findings

reveal that the most informative samples, such as those located near the decision boundary and-or associated with high prediction uncertainty, did not yield the best-optimal performance. Instead, the most confidently predicted samples, which are typically less informative in traditional active learning, consistently led to higher-classification-accuracy-superior decoding performance. This observation aligns-with-supports our hypothesis that high-uncertainty examples may introduce ambiguity and provide less-effective-suboptimal guidance during inference. In contrast, prototypical and easily classifiable examples better-represent-the-underlying-characteristics-of-each-class-more-faithfully-capture-the-intrinsic-characteristics-of-their-respective-classes, thereby supporting more accurate model predictions.

Based-on-Motivated-by this insight, we propose Purity-based Unique-Round-Example-selection (PURE)PGAP, a simple yet effective pipeline for selecting in-context examples in EEG-based tasks. For each training sample, we compute its k nearest neighbors and calculate the proportion of neighbors sharing that share the same class label—a metric we refer to as local same-label purity. As illustrated in Algorithm ??1, all training samples are ranked based-on-by their purity scores, and the top- n most “purePGAP” samples are selected as in-context examples. This strategy-not-only-selection-strategy favors the most representative samples within each class but also-inherently-avoids-outliers-as-samples-while-inherently filtering-out-outliers, which tend to be surrounded by mixed-class neighbors and thus receive lower purity scoresand-are less-likely-to-be-chosen.

Although PURE-PGAP does not explicitly enforce diversity, we empirically observe that augmenting it with greedy diversity sampling leads to performance degradation. This indicates that, in the context of LLM-based EEG classificationdecoding, purity alone is-serves-as a more effective selection criterion for example selection.

PURE-PGAP adopts a one-time selection strategy, in which a fixed set of in-context examples is selected once from the training set and reused across all test instances. Compared to conventional per-instance selection, this approach offers significant computational advantagesbenefits:

- 1) *Reduced-Reducing computational overhead:* -It avoids repeated-example-PGAP eliminates the need for dynamic retrieval by avoiding repeated exemplar selection for each test input, eliminating the need for instance-wise retrievalinstance.
- 2) *Improved-Improving batch efficiency:* -Since all test samples share the same in-context examples, multiple test inputs can be batched into a single prompt, enabling parallel predictions with a single API call. In contrast, per-instance selection requires constructing-a-unique prompt-for-each-test-input, yielding-individualized prompt construction, allowing only one prediction per query.

C. Prompt Design

The final step involves constructing the in-context prompt by combining the selected examples and their corresponding labels together with the task description, contextual information,

and explicit output instructions. This prompt is then presented to the LLM alongside the test input.

Our design follows established best practices in prompt engineering to ensure clear and effective communication with the LLM [48]. Specifically, the task description is kept-concise, unambiguous, and placed-positioned at the beginning of the prompt to set-explicitly-define the model’s objective. The context section includes-incorporates relevant background information and a few representative input-output pairs, enabling the model to infer the desired-underlying mapping pattern from demonstrations.

Explicit-output-instructions-are-also-incorporated-to-To further guide the model toward producing structured-and-predictable-responses, which-are-critical-for-downstream evaluation-and-integrationconsistent-and-structured-outputs, explicit-output-instructions-are-incorporated. These instructions specify both content constraints (e.g., label formats, output options) and style constraints (e.g., use of quotation marks, line breaks).-In-addition,we-apply-, which-are-critical-for-downstream-evaluation-and-integration. Additionally, we employ prompt formatting techniques such as section delimiters (e.g., “###” or newline markers), consistent-standardized indentation, and token normalization to delineate boundaries between different components—such-as-, including descriptions, examples, and the query—thereby-improving-input-, thereby-improving-prompt clarity and model interpretability. A typical prompt structure used in our study is showed-shown in Fig 3.

To avoid prompt truncation or-and exceeding token limits, all components are carefully optimized for brevity without compromising-informativenesswhile preserving essential information. This balance is particularly important-when-working-critical-when-interacting with token-limited APIs or processing long test inputs. The final prompt format is compatible with both open-source and API-based LLMs and is designed to be-reusableacross-batches-of-test-support reusable, batch-level inputs for efficient inference.

IV. DATASETS EXPERIMENTS AND RESULTS

This section introduces five-

A. Datasets

Five MI datasets, one sleep dataset, and one epilepsy dataset used-are-utilized in our experiments.

B. MI Datasets

1) *MI Datasets:* All MI datasets used in our experiments are binary classification tasks and were obtained using the MOABB framework [49]. Specifically, two datasets from BCI Competition IV (BNCI2014001 and BNCI2014004) [50] contain left-vs-right hand imagery, while the other three: BNCI2014002, BNCI2015001, and Weibo2014, contain right-hand vs. feet imagery.

In each session, a subject sat comfortably in an armchair positioned in front of a computer screen. At the beginning of each trial, a fixation cross appeared on a black background,

Algorithm 1: ~~Purity-based Unique Round Example Selection (PURE)PGAP~~

Input: N labeled samples, $\{\mathbf{x}_n\}_{n=1}^N$, where $\mathbf{x}_n \in \mathbf{R}^d$;
 C , number of classes;
 M , number of samples to select;
 K , neighborhood size

Output: A list of selected sample indices with balanced class distribution L .

Ensure M is divisible by C ;

foreach sample $n = 1$ to N **do**

Compute distances between $\mathbf{x}_n$ and all other samples;

Find K nearest neighbors of $\mathbf{x}_n$ (excluding itself);

Compute same-label purity score:

$$\text{purity}(\mathbf{x}_n) = \frac{\text{Number of same-label neighbors}}{K}$$

end

Group all samples by class label;

Sort each group in descending order of purity score;

foreach class $c = 1$ to C **do**

Select top M/C samples with highest purity;

Add their indices to output list L ;

end

return Selected sample indices list L .

Task Description

Given a set of feature vectors of EEG samples, determine whether the category of the sample to be predicted is A or B.

Demonstrations

Features: [0.23, 1.45, 0.87, ..., 0.91]

Label: A

Features: [1.14, 0.76, 1.09, ..., 0.67]

Label: B

...

Samples to be predicted

Features: [0.41, 1.30, 0.99, ..., 0.80]

Features: [1.08, 0.64, 1.05, ..., 0.72]

Features: [0.27, 1.38, 0.90, ..., 0.76]

Features: [0.37, 4.38, 2.90, ..., 0.16]

...

Instructions

Predict the label of the following input and return in a JSON list.

Fig. 3. A typical prompt structure used in our experiments.

accompanied by a short auditory warning tone. After a brief delay, a directional cue in the form of an arrow pointing left, right, down, or up appeared on the screen, corresponding to one of the four MI classes: left hand, right hand, feet, or tongue. Subject performed the imagery task until the fixation cross disappeared. Each trial ended with a brief black screen before the next trial began [50].

Notably, for BNCI2015001, the last 3 subjects were excluded due to poor performance [51]. And for Weibo2014, only right-hand and feet trials were used for binary classification in this study. From the original 60 usable channels, we selected 21 MI-related channels: FC3, FC1, FCz, FC2, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP1, CPz, CP2, CP4, P3, P1, P2, and P4.

B. Sleep-Datasets

1) Sleep Dataset: The Sleep-EDF Expanded (Sleep-EDFx) dataset is intended for sleep research, particularly for sleep stage classification. It contains 197 full-night polysomnographic (PSG) recordings, including EEG (Fpz-Cz and Pz-Oz), EOG, EMG, and event markers. Corresponding hypnograms (sleep stage annotations) were manually scored by trained technicians following the Rechtschaffen and Kales manual and are provided alongside the recordings.

Only EEG channels are used for a five-stage sleep classification task. Due to the large volume of data, we selected recordings from first seven subjects, using one night of data per subject.

During the signal processing and feature extraction stages, EEG data from the Sleep-EDFx dataset were first bandpass

filtered between 0.5 Hz and 35 Hz. Since the recordings span an entire night and include prolonged wake periods at the beginning and end, only the first and last 30 minutes of wakefulness were retained to reduce redundancy. Sleep stages were labeled according to the AASM standard and grouped into five classes: Wake (W), N1, N2, N3, and REM. Each trial was segmented into non-overlapping 30-second epochs.

For each epoch, the signal from every channel was processed individually. The PSD was computed using the Welch algorithm. We then defined five canonical frequency bands: delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), spindle (12-14 Hz) and beta (14-30 Hz). For each band, the mean PSD within the corresponding frequency range was calculated by averaging the PSD values across that band. This process was repeated across all channels and frequency bands, resulting in a feature vector of size $C \times B$ per epoch, where C is the number of EEG channels and B is the number of frequency bands.

B. Epilepsy-Datasets

1) Epilepsy Dataset: The CHB-MIT Scalp EEG Database comprises EEG recordings from 22 pediatric subjects (5 males, aged 3–22 years, and 17 females, aged 1.5–19 years) with intractable seizures. Subjects were monitored over multiple days following the withdrawal of anti-seizure medication to capture seizure activity and evaluate their suitability for surgical intervention. In total, 182 seizures were annotated with precise onset and offset timings.

All signals were sampled at 256 Hz with 16-bit resolution. Most recordings contain 23 EEG channels, though a few have

TABLE I
SUMMARY OF THE SEVEN EEG DATASETS.

BCI Paradigm	Dataset	Number of Subjects	Number of Channels	Sampling Rate (Hz)	Trial Length (seconds)	Number of Trials per Subject	Class Labels
MI	BNCI2014001	9	22	250	4	288	left hand, right hand
	BNCI2014004	9	3	250	4.5	680-760	left hand, right hand
	BNCI2014002	14	15	512	5	160	right hand, both feet
	BNCI2015001	12(9)	13	512	5	400/600	right hand, both feet
	Weibo2014	10	60(22)	200	4	140, 160	right hand, both feet
Sleep	Sleep-EdfX	7	2	100	30	1025	W, N1, N2, N3, REM
Epilepsy	CHB-MIT	10	23	256	3	76-618	seizures, non-seizures <u>seizure, non-seizure</u>

24 or 26. Electrode placement follows the International 10-20 system for EEG acquisition.

~~We selected the first ten subjects for~~ Considering the computational cost, we only used the first nine subjects in our experiments. The raw EEG signals were segmented using non-overlapping sliding windows of 3 seconds. We extracted all seizure segments and randomly selected an equal number of non-seizure segments from the remaining data to ensure class balance.

For feature extraction, we employed a commonly used and effective time-domain features for seizure detection: band power. Specifically, the raw EEG signals were filtered into five standard frequency bands: δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz), and γ (30–50 Hz).

For each frequency band, the signal from every channel was individually filtered and the power was calculated as the mean squared amplitude over time. The power values from all channels were then averaged to obtain a single scalar representing the overall energy in that frequency band. This process was repeated for all five bands, resulting in a five-dimensional feature vector per EEG epoch. These band power features provide a compact summary of the frequency-specific signal energy across the scalp.

V. EXPERIMENTS

A. Compared Approaches

We compare PGAP with various existing approaches:

- 1) BL, which randomly selects all M samples.
- 2) CTD, which applies the k -means algorithm within each class and selects centroids.
- 3) RD [37], which starts by selecting the centroid of each class as the initial point. Then, iteratively performs additional rounds of clustering on each class and selects the centroid of the largest cluster not yet covered by the previously selected points.
- 4) SVM-near, which trains a SVM classifier on the training set and selects, from each class, the sample closest to the decision boundary.
- 5) SVM-far, which in contrast to SVM-near, selects the sample farthest to the decision boundary within each class. To mitigate the impact of outliers, SVM-far further filters candidates by checking the proportion of same-label samples among their five nearest neighbors.
- 6) PGAP, which selects the highest-purity samples from each class as describe in Section III-B.

~~This section conducts experiments to demonstrate the effectiveness of PURE. We use~~

B. Implementation

We utilize the Qwen-2.5-14B-Instruct-1M model as the LLM, and apply two demonstrations per category, resulting in a total of four ICL demonstrations for the MI binary classification dataset, ten for the sleep five-class classification task, and four for the epilepsy binary classification dataset.

C. Set-up

We conducted experiments under three settings. For the MI datasets, we adopted (i) within-subject, cross-session evaluation and (ii) cross-subject evaluation. For the sleep and epilepsy dataset, as only one session per subject is available, we employed (iii) within-subject five-fold cross-validation instead of (i). For the epilepsy dataset, due to the large variation in the number of seizure segments across subjects, cross-subject experimental results may be affected. Therefore, we only conduct within-subject experiments.

All ~~methods~~ approaches were run 5 times. Random selection baselines were repeated 20 times. For five-fold cross-validation, the entire process was repeated 5 times, totaling 25 runs ~~per method~~ for each approach. Classification accuracy was utilized as the evaluation metric.

- 1) *Within-Subject Cross-Session*: Datasets were partitioned into non-overlapping training and testing sets. Specifically:

- BNCI2014001: One session for training and another for testing, each containing 144 trials.
- BNCI2014004: First 3 sessions for training; last 2 for testing.
- BNCI2014002: First 5 runs for training; last 3 for testing. Each run contains 10 right-hand and 10 feet trials.
- BNCI2015001: One session for training and one for testing (400 trials total). Subjects 8–9 use first 2 sessions for training and the third for testing.
- Weibo2014: First half of the session for training; second half for testing.

- 2) *Within-Subject five-Fold Cross-Validation*: Trials were shuffled and split into 5 folds with class balance preserved.

- 3) *Cross-Subject*: Leave-one-subject-out. Remaining subjects used for training. Euclidean Alignment (EA) [5] was applied to each subject's data.

C. Main Results

~~We compare PURE with various existing algorithms:-~~

- 1) ~~BL, which randomly selects all M samples.-~~
- 2) ~~CTD, which applies the k -means algorithm within each class and selects centroids.-~~
- 3) ~~RD [37], which starts by selecting the centroid of each class as the initial point. Then, iteratively performs additional rounds of clustering on each class and selects the centroid of the largest cluster not yet covered by the previously selected points.-~~
- 4) ~~SVM-near, which trains a SVM classifier on the training set and selects, from each class, the sample closest to the decision boundary.-~~
- 5) ~~SVM-far, which in contrast to the above method, selects the sample farthest to the decision boundary within each class. To mitigate the impact of outliers, the method further filters candidates by checking the proportion of same-label samples among their five nearest neighbors.-~~
- 6) ~~PURE, which selects the highest-purity samples from each class as describe in section ??-~~

Tables II- VIII presents the classification accuracies on the seven EEG datasets, respectively. We can observe that:

- 1) Active selection of ICL demonstrations significantly outperforms random selection. In both within-subject and cross-subject settings, actively selected demonstrations consistently lead to a better prompting effect on the LLM, resulting in higher prediction accuracy and more stable outputs compared to random selection.
- 2) High-informativeness samples yield the worst prompting effect in ICL. Contrary to conventional expectations in traditional machine learning, selecting the most informative samples leads to an inferior prompting effect on the LLM, even worse than random selection. This suggests a fundamental difference between how LLMs and traditional classifiers benefit from informative instances.
- 3) Low-informativeness samples achieve the best prompting effect. Demonstrations with low informativeness and high model confidence, as exemplified by SVM-far and **PURE-PGAP**, result in superior prompting effects. We hypothesize that when classifying complex numerical data, LLMs primarily rely on statistical distributional patterns rather than constructing deep decision boundaries. Therefore, more prototypical and easily distinguishable examples are more effective in guiding the model, whereas highly ambiguous samples fail to provide meaningful distinctions.
- 4) **PURE-PGAP** consistently outperforms SVM-far. Although SVM-far has been optimized to avoid outlier selection and shows competitive performance, **PURE-PGAP** still achieves better overall prompting effects across datasets and experimental conditions.
- 5) **PURE-PGAP** achieves consistent superiority across multiple datasets and paradigms. ~~Our proposed method~~

~~PURE~~-**The proposed PGAP** consistently outperforms random selection across all experiments and achieves optimal results in the majority of cases. Specifically, in the within-subject setting, **PURE-PGAP** improves over random selection by 8.04%, 8.19%, 6.03%, 3.43%, and 4.09% across the five MI datasets, by 5.44% on the sleep dataset, and by 3.68% on the epilepsy dataset. In the cross-subject setting, **PURE-PGAP** achieves similar improvements: 10.17%, 6.32%, 5.35%, 1.37%, and 1.93% across the five MI datasets, 5.67% on the sleep dataset.

- 6) Representativeness- and diversity-based ~~methods~~ **approaches** (RD and CTD) also outperform random selection. Although RD and CTD perform slightly worse than **PURE-PGAP** overall, they consistently outperform random selection across all experiments. Notably, in cross-subject settings, their performance significantly improves, achieving results comparable to or even exceeding **PURE-PGAP** on several MI datasets. Moreover, since these ~~methods~~ **approaches** do not strictly rely on label information (labels are only used to maintain class balance), they exhibit broader applicability to scenarios where label information is limited or unavailable.

D. Analysis Effect of the Number of Demonstration Samples

In the preceding experiments, we selected two samples per category as demonstrations. We investigated the impact of varying the number of demonstration samples on performance. The experiments are conducted on the BNCI2014001 dataset. Figure 4 presents the performance trends as the number of demonstration samples increases, from which we observe the following:

- 1) The performance of the LLM improves as the number of demonstration samples increases.
- 2) As the number of demonstration samples increases, the marginal performance gains progressively decrease, and the results across different selection ~~methods~~ **approaches** tend to converge.
- 3) The results of random selection exhibit large fluctuations across repeated trials, indicating significant instability. This further reflects the sensitivity of LLMs to the choice of demonstrations.
- 4) **PURE-PGAP** consistently outperforms random and other selection ~~methods~~ **approaches** across different numbers of demonstrations, and exhibits the smallest variance across repeated experiments, indicating the highest stability.

E. Comparison of Different Models LLMs

To further assess the generalizability of the proposed ~~method~~ **PGAP**, we evaluated its performance on alternative LLMs. Specifically, DeepSeek-V3 was selected for comparative analysis. Experiments were conducted on the BNCI2015-002 dataset, and the corresponding results are presented in Table IX. The following observations can be made:

- 1) Active selection ~~methods~~ **approaches** consistently outperform random selection across different LLMs.

TABLE II

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2014001. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification										
Methods - Approaches	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	49.31	49.03	67.64	56.11	48.19	55.28	52.22	56.81	83.47	57.56
RD	55.42	48.89	77.92	64.44	48.06	53.19	54.31	71.11	80.14	61.50
CTD	56.81	44.17	71.53	62.92	50.28	51.81	54.58	76.25	86.11	61.61
SVM-near	38.89	50.00	45.00	53.06	50.56	47.36	48.19	46.11	71.53	50.08
SVM-far	58.89	47.64	75.00	61.11	52.78	55.69	57.22	73.06	85.00	62.93
PURE - PGAP	67.08	51.94	68.61	68.75	52.50	59.44	58.19	77.22	86.67	65.60 (↑8.04)
Cross-subject classification										
BL	48.68	50.42	66.25	59.93	53.33	55.83	45.69	53.68	38.54	52.48
RD	61.25	51.18	74.44	64.65	57.50	61.81	67.64	70.69	52.85	<u>62.45</u>
CTD	58.13	50.14	63.54	63.82	56.53	54.03	63.47	74.44	76.53	62.29
SVM-near	37.15	47.50	52.99	59.65	48.82	43.26	36.11	42.43	26.67	43.84
SVM-far	60.76	47.85	66.67	61.18	54.37	61.53	58.75	72.08	74.10	61.92
PURE - PGAP	60.76	52.15	70.07	60.76	54.79	62.92	62.64	68.89	70.83	62.65 (↑10.17)

TABLE III

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2014002. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification															
Methods - Approaches	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	Avg.
BL	61.33	75.00	72.67	60.67	59.33	68.33	84.00	64.00	88.00	54.67	47.67	57.00	47.33	52.67	63.76
RD	65.00	74.67	76.67	68.00	60.67	84.00	89.67	71.67	88.67	56.33	73.67	47.33	49.00	57.67	68.79
CTD	66.67	78.67	80.33	61.33	66.00	83.00	90.00	51.67	94.33	56.00	67.00	54.33	45.33	54.00	67.76
SVM-near	37.33	47.67	20.67	47.67	41.00	89.00	59.67	60.00	72.33	53.33	81.33	61.67	44.33	56.67	55.19
SVM-far	71.00	69.33	77.33	75.33	64.67	82.33	86.67	70.00	92.33	54.00	81.00	61.33	50.33	52.33	<u>70.57</u>
PURE - PGAP	63.33	74.67	88.33	77.67	61.67	83.00	88.67	71.67	90.00	55.00	79.00	70.00	46.33	58.00	71.95 (↑8.19)
Cross-subject classification															
BL	55.50	60.50	60.62	56.75	55.88	62.38	60.63	62.63	80.25	61.88	59.50	55.75	56.13	48.75	59.80
RD	56.63	69.00	86.75	61.37	67.50	71.88	85.50	70.37	61.25	64.50	61.75	65.00	57.25	44.75	65.96
CTD	55.63	68.75	85.88	63.00	66.63	73.00	85.25	67.00	82.25	65.25	62.25	67.62	55.50	48.88	67.64 (↑7.84)
SVM-near	56.50	29.38	26.75	45.88	40.00	67.37	32.00	66.87	36.75	45.50	63.87	38.12	44.12	53.25	46.17
SVM-far	56.50	64.50	81.75	57.13	59.12	67.87	81.75	61.50	83.63	56.37	62.50	60.88	53.75	53.38	64.33
PURE - PGAP	57.38	66.25	86.75	55.13	62.75	69.12	85.00	62.38	85.13	60.75	65.50	66.25	51.38	51.88	<u>66.12</u>

TABLE IV

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2014004. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification										
Methods - Approaches	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	51.44	55.00	51.00	65.31	54.94	47.62	50.44	58.94	51.31	54.00
RD	55.44	49.57	55.40	75.25	54.00	62.62	52.81	71.50	52.81	58.82
CTD	49.19	56.93	52.87	65.81	50.00	52.19	54.88	56.94	54.31	54.79
SVM-near	49.31	56.93	51.40	47.25	51.25	45.94	42.87	52.75	42.81	48.95
SVM-far	53.25	56.29	54.00	77.25	55.50	54.69	49.75	75.94	57.06	59.30
PURE - PGAP	51.67	50.14	53.67	76.88	61.44	57.25	52.87	78.25	58.13	60.03 (↑6.03)
Cross-subject classification										
BL	50.92	49.79	50.56	54.73	51.84	49.83	51.53	52.13	54.44	51.75
RD	54.89	52.35	50.44	64.81	50.19	57.50	50.14	56.76	48.17	53.92
CTD	47.00	48.26	50.11	45.08	46.73	51.61	49.72	46.76	52.56	48.65
SVM-near	45.36	48.85	49.14	41.84	47.35	48.33	48.17	59.89	48.67	48.62
SVM-far	59.11	52.88	50.00	68.27	53.14	55.19	52.00	61.47	52.36	<u>56.05</u>
PURE - PGAP	57.00	52.56	52.64	69.05	55.51	57.14	53.25	64.76	52.03	57.10 (↑5.35)

TABLE V

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2015001. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification										
Methods -Approaches	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	74.30	88.30	70.50	82.10	65.00	51.30	76.80	55.00	58.40	69.08
RD	79.60	89.00	55.90	77.40	70.40	54.90	76.50	58.30	54.30	68.48
CTD	79.00	89.60	55.50	78.50	69.30	56.00	74.50	58.30	50.50	67.91
SVM-far	67.00	86.30	83.00	82.60	67.70	52.20	81.70	64.90	62.40	<u>71.98</u>
SVM-near	66.70	60.80	53.00	28.60	40.80	50.40	41.20	45.80	51.60	48.77
PURE -PGAP	71.80	88.80	70.80	82.80	74.40	51.20	76.00	69.20	67.60	72.51 (↑3.43)
Cross-subject classification										
Methods -Approaches	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	70.25	80.15	68.45	65.55	61.85	56.40	55.35	56.60	64.03	64.29
RD	78.50	83.55	65.85	55.60	68.75	56.65	62.00	56.63	63.27	<u>65.64</u>
CTD	71.90	79.35	66.85	62.70	68.35	55.75	58.50	57.87	63.77	65.00
SVM-far	76.15	76.80	66.00	66.20	62.70	50.15	56.70	56.67	58.67	63.34
SVM-near	73.95	69.45	55.25	54.40	69.00	57.45	57.10	56.77	60.03	61.49
PURE -PGAP	76.70	85.30	66.15	66.45	63.50	56.25	60.50	53.70	62.40	65.66 (↑1.37)

TABLE VI

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON WEIBO2014. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification											
Methods -Approaches	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Average
BL	67.50	47.50	70.25	53.00	66.50	68.29	56.25	88.50	84.50	64.50	66.68
RD	67.75	46.50	73.00	59.25	76.50	68.57	63.75	84.00	84.75	71.75	69.58
CTD	74.75	49.50	71.50	54.00	69.75	69.71	55.75	84.00	83.50	67.25	67.97
SVM-far	76.25	59.00	73.50	48.50	67.50	75.43	61.50	77.75	90.00	69.25	<u>69.87</u>
SVM-near	37.75	45.75	60.25	46.25	44.25	49.14	43.50	43.75	88.75	63.00	52.24
PURE -PGAP	69.50	61.75	76.00	67.50	69.25	71.71	57.25	80.00	87.50	67.25	70.77 (↑4.09)
Cross-subject classification											
Methods -Approaches	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Average
BL	63.69	54.06	62.32	52.19	62.94	63.07	73.07	78.25	78.25	61.50	64.93
RD	71.13	52.37	66.87	51.38	64.50	64.14	75.75	80.37	87.75	64.87	67.91 (↑2.98)
CTD	72.94	48.75	66.75	51.88	62.25	55.86	77.62	63.75	85.75	66.25	65.18
SVM-far	68.37	54.13	72.12	57.87	64.12	63.29	71.12	80.50	83.37	60.38	<u>67.53</u>
SVM-near	63.00	53.75	35.88	46.88	54.87	68.71	58.25	43.00	26.37	60.38	51.11
PURE -PGAP	69.75	55.13	71.37	55.13	60.50	60.00	73.50	74.25	89.38	59.50	66.86

TABLE VII

WITHIN-SUBJECT AND CROSS-SUBJECT CLASSIFICATION ACCURACIES (%) ON SLEEP-EDF. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Within-subject classification								
Methods -Approaches	S1	S2	S3	S4	S5	S6	S7	Average
BL	29.81	30.93	31.93	28.26	35.18	30.18	31.87	31.17
RD	26.31	28.76	36.58	29.13	34.81	32.76	35.06	31.92
CTD	23.92	27.08	34.76	26.28	35.66	29.38	28.30	29.34
SVM-far	37.83	28.43	36.49	31.51	39.63	39.25	37.95	<u>35.87</u>
SVM-near	28.65	25.85	27.47	26.07	31.34	31.32	26.40	28.16
PURE -PGAP	35.57	33.93	35.71	35.97	50.78	31.48	32.80	36.61 (↑5.44)
Cross-subject classification								
BL	23.59	21.81	19.52	24.87	27.20	20.36	23.18	22.93
RD	22.67	24.41	17.25	18.27	22.47	18.01	22.15	20.89
CTD	21.49	20.47	19.62	18.79	19.52	18.10	25.96	20.85
SVM-far	24.81	23.24	17.98	19.89	22.68	16.16	32.36	22.73
SVM-near	20.44	21.56	16.13	27.30	22.83	30.13	23.01	<u>23.06</u>
PURE -PGAP	29.79	25.81	26.28	20.37	34.64	34.66	24.63	28.60 (↑5.67)

TABLE VIII

WITHIN-SUBJECT CLASSIFICATION ACCURACIES (%) ON CHB-MIT. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Methods Approaches	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	84.93	84.35	81.25	59.68	84.52	51.85	82.82	74.88	90.43	77.12
CTD	71.00	73.22	79.27	41.59	87.44	48.67	85.18	79.29	90.71	71.45
RD	55.40	70.44	78.41	44.27	87.02	51.07	83.64	74.27	88.71	70.43
SVM-near	39.87	68.53	73.33	59.88	79.08	42.17	79.91	67.99	89.57	66.87
SVM-far	81.33	83.70	79.15	74.16	81.10	58.37	80.00	74.39	89.57	<u>77.35</u>
PURE PGAP	85.40	86.03	82.98	72.64	86.17	64.20	83.55	76.27	89.43	80.80 (↑3.68)

TABLE IX

WITHIN-SUBJECT CLASSIFICATION ACCURACIES (%) ON BNCI2015001 USING THE DEEPSEEK-V3 MODEL. THE BEST PERFORMANCE IN EACH SUBJECT IS MARKED IN BOLD, AND THE SECOND BEST UNDERLINED.

Methods Approaches	S1	S2	S3	S4	S5	S6	S7	S8	S9	Average
BL	87.40	92.90	95.30	84.10	79.90	63.30	76.30	59.20	71.30	78.86
RD	93.80	91.20	79.50	79.10	80.40	59.80	73.20	56.40	76.10	76.61
CTD	93.10	93.20	80.00	76.90	80.10	62.70	71.80	57.30	87.50	78.10
SVM-near	34.30	39.60	33.90	11.10	33.50	62.10	38.50	23.90	38.50	35.04
SVM-far	75.30	88.20	85.00	71.00	76.50	60.70	61.10	58.10	84.60	73.40
PURE PGAP	75.40	93.00	91.20	81.50	86.40	61.10	80.30	64.10	86.40	79.93 (↑1.07)

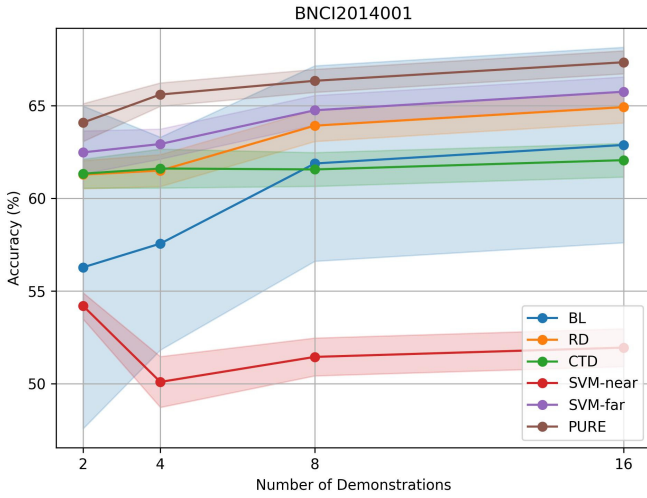


Fig. 4. Performance under different numbers of in-context demonstrations on BNCI2014001 datasets using BL, RD, CTD, SVM-near, SVM-far and ~~PURE~~~~PGAP~~ approaches. The LLM used here is our base model: Qwen-2.5-14b-instruct-1m.

- 2) The relative performance hierarchy among all ~~methods~~~~approaches~~ remains consistent with previous experiments, confirming the robustness of active selection strategies across different model architectures.
- 3) Employing larger and more powerful models leads to a consistent improvement in accuracy for all ~~methods~~~~approaches~~. Remarkably, on certain datasets, the results surpass the SOTA performance achieved by conventional deep neural networks, highlighting the considerable potential of LLM-based ICL few-shot learning approaches in non-text domains such as EEG signal analysis.

V. CONCLUSIONS

This paper proposed a purity-guided ~~unique-round-example~~ (~~PURE~~) ~~approach for EEG signal classification using active prompting (PGAP) framework for EEG decoding with LLMs~~ in a few-shot in-context learning scenario. Existing demonstration selection approaches commonly used in NLP and CV domains are often ineffective for EEG data due to its complexity and inherent lack of semantic interpretability. ~~PURE~~~~PGAP~~ addresses these limitations by combining domain-specific feature extraction with an active selection strategy based on local same-label purity, selecting compact sets of representative, low-entropy examples. Experiments on multiple EEG datasets across diverse BCI paradigms show that ~~PURE~~~~PGAP~~ consistently outperforms random and baseline selection strategies.

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